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18 September 1969
SUPERSEDING
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MILITARY SPECIFICATION

LAUNCHING GROUP MARK 16 MOD 4

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification establishes the requirements for the Launching Group Mark 16 Mod 4, Naval Ordnance Systems Command LD 272606, less Launcher and Missile Simulator Mark 6. The launching group stows and launches rocket-thrown torpedoes and rocket-thrown depth charges from naval vessels. The Launching Group Mark 16 Mod 4 is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-23957	Power Drive, Train, Mk 61 Mod 2 Power Drive, Elevation, Mk 62 Mod 2 Chassis, Amplifier, LD 272613

DRAWINGS

Naval Ordnance Systems Command

LD 272569	Dummy Director Mk 3 Mod 8 and Error Recorder Mk 12
LD 272606	Launching Group Mark 16 Mod 4
LD 490381	Boresight Equipment

FSC 1055

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DRAWINGS

Naval Ordnance Systems Command (Continued)

LD 497791	Rail Gage
298871-1	Angle Pinch Bar (Training Circle and Mount Bearing Subassembly Inspection)
1453351	Test Round
1656467	Test Cam (Switch Setting, Firing Interrupter)
1735447	Guide, Launcher Mark 7 Mod 1
1806247	Restraining Lug, Missile
1877495	Launcher Mark 112 Mod 2
SA 2357954	Test Fixture (Elevation Buffer)
SA 2357955	Hydro Test Fixture (Train Buffer)
SA 2369575	Test Fixture (Missile Restraining Latch)
SA 2371509	High Potential Test Kit
SA 2372138	Test Fixture
SA 2372270	Target and Stand
SA 2372685	Test Cam (Switch Setting, Firing Interrupter)
SA 2373409	Power Distribution Panel (Wiring Schematic)
SA 2373586	Test Synchro (Wiring Schematic)
SA 2373587	Test Panel (UB Plot Simulator Wiring Schematic)
SA 2373588	Test Set (Missile Simulator Wiring Schematic)
SA 2490553	High Potential Test Set

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PUBLICATIONS

Naval Ordnance Systems Command

OD 13040	Launching Group Mk 16 Mod 4, Factory Acceptance Test Rev. 3 - Serial Nos. 219 - 256 Rev. 4 - Serial Nos. 257 - 328 Rev. 5 - Serial Nos. 329 and up
OD 13041	Launching Group Mk 16 Preservation, Packaging, Packing, Shipping, Handling and Storage Instructions

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publications

Handbook H28 and all Supplements thereto	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

2.3 Publications for information only -

Naval Ordnance Systems Command Publications

OP 2385, Vol. 1	Launching Group Mk 16; Launcher Mk 112, Power Control Panel Mk 198, Launcher Captain's Control Panel Mk 199, Description, Operation and Maintenance
OP 2385, Vol. 2	Launching Group Mk 16; Train Power Drive Mk 61, Elevation Power Drive Mk 62, Power Drive Amplifier, Mk 153 Amplifier Chassis, Description and Maintenance

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Naval Ordnance Systems Command Publications (Continued)

OP 2385, Vol. 3	Launching Group Mk 16; Launcher and Missile Simulator Mark 6, Description, Operation and Maintenance
NAVORD OD 10729	Launching Group Mk 16 Mod 4 Shipboard Installation and Checkout

3. REQUIREMENTS

3.1 Description - The Launching Group Mk 16 Mod 4, described herein, is designed to stow and launch the Rocket Thrown Depth Charge Mk 2 and the Rocket Thrown Torpedo Mk 3, and shall be composed of the following components:

Launcher Mk 112 Mod 2

Stand Mk 107 Mod 2

Carriage Mk 7 Mod 2
Train Power Drive Mk 61 Mod 2
Elevation Power Drive Mk 62 Mod 2

Guide Mk 7 Mod 1

Launcher Captain's Control Panel Mk 199 Mod 3

Launcher and Missile Simulator Mk 6 Mod 2

3.2 General requirements -

3.2.1 Materials - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the Launching Group Mk 16 Mod 4 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 272606.

3.2.2 Finishes, coating, construction and dimensions - Finishes, coating, construction and dimensions for the components of the Launching Group Mk 16 Mod 4 shall be strictly in accordance with the applicable drawings, specifications, and publications listed in LD 272606.

3.2.3 Moisture, fungus and corrosion - The Launching Group Mk 16 Mod 4 shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified on the drawings and specifications in LD 272606.

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3.2.4 Marking - Marking of assemblies, subassemblies and parts shall be in accordance with the applicable drawings listed in LD 272606.

3.2.5 Interchangeability - Individual parts of assemblies, fabricated in agreement with their applicable drawings and specifications, shall be interchangeable with mating parts of the unit and other like units. If deviations greater or smaller than the tolerances shown on the drawings or specifications are required for interchangeability; or if hand fitting beyond that shown on the applicable drawings is required; then changes in dimensions are necessary and revision and correction of the applicable drawings is required. See 3.2.9.

3.2.6 Gages - The supplier shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Naval Final Inspection Gages for guidance in the design of the supplier's inspection gages. However, such procedure shall not serve to relieve the supplier of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Naval Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and all Supplements thereto. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the Supplement to the aforementioned Handbook H28.

3.2.8 Requests for deviation - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be submitted to the procuring activity. The deviation shall have been authorized by the Naval Ordnance Systems Command or by a Naval activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The request shall define the suggested deviation in sufficient detail to permit the procuring activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated.

3.2.9 Conflicting requirements - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring activity or appointed agent for interpretation, clarification and resolution (or correction).

3.3 Performance requirements and product characteristics - The Launching Group Mk 16 Mod 4 shall meet the following performance requirements and product characteristics:

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3.3.1 Limits of operation - The limits of operation with respect to the stow position, shall be:

TRAIN	
Right:	179 degrees, plus or minus 9 minutes
Left:	179 degrees, plus or minus 9 minutes
ELEVATION	
Elevating:	7,100 minutes, plus or minus 15 minutes
Depressing:	1,850 minutes, plus or minus 15 minutes

Base reference for stow position shall be:

TRAIN:	Zero degrees
ELEVATION:	2,000 minutes

3.3.2 Velocity -

3.3.2.1 Train velocity -

3.3.2.1.1 Lowest maximum velocity - The lowest maximum velocity in both directions of train shall be 38 degrees per second when operating with all cells loaded with test rounds, Drawing 1453351, and any guide elevated to 6,980 minutes.

3.3.2.1.2 Maximum velocity - The launcher shall not exceed 45 degrees per second in either direction of train when operating with all guides in stow position and one inboard cell loaded with test round, Drawing 1453351.

3.3.2.2 Elevation velocity -

3.3.2.2.1 Lowest maximum velocity - The lowest maximum velocity in both directions of elevation of a guide shall be 24 degrees per second when operating with both cells of guide loaded with test rounds, Drawing 1453351.

3.3.2.2.2 Maximum velocity - A guide shall not exceed 26 degrees per second in either direction in elevation when operating with the lower cell of guide loaded with test round, Drawing 1453351.

3.3.3 Acceleration and deceleration -

3.3.3.1 Train acceleration and deceleration -

3.3.3.1.1 Lowest maximum acceleration and deceleration - The lowest maximum acceleration and deceleration in both directions of train shall be 55 degrees per second when operating with all cells loaded with test rounds, Drawing 1453351, and any guide elevated to 6,980 minutes.

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3.3.3.1.2 Maximum acceleration - The launcher shall not exceed 150 degrees per second per second in either direction of train when operating with all guides in stow position and one inboard cell loaded with test round, Drawing 1453351.

3.3.3.2 Elevation acceleration and deceleration -

3.3.3.2.1 Lowest maximum acceleration and deceleration - The lowest maximum acceleration and deceleration in both directions of elevation shall be 85 degrees per second per second when operating with both cells of guide loaded with test rounds, Drawing 1453351.

3.3.3.2.2 Maximum acceleration - The launcher shall not exceed 250 degrees per second per second in either direction of elevation when operating with the lower cell of guide loaded with test round, Drawing 1453351.

3.3.4 Mechanical-hydraulic balance - Voltage across the terminals of the torque motor, for both train and elevation power drives, shall be plus or minus 2 volts DC when the power drives are operating in synchronism with the director and the temperature of the hydraulic fluid at the pump case drain is 120 degrees to 135 degrees F.

3.3.5 Mechanical limit stops - The mechanical limit stops in the receiver-regulators of the train and elevation power drives shall decelerate the launcher to rest within the following limits when loaded as specified in 3.3.3.1.1 and 3.3.3.2.1.

TRAIN:	150 degrees per second per second
ELEVATION:	250 degrees per second per second

3.3.6 Static accuracy of correspondence - The position of the launcher shall correspond to the position indicated on the synchro dials of the director, within two minutes, independent of hunt, as read at the receiver-regulator dials.

3.3.7 Hunt - With the launcher loaded as specified in 3.3.2.1.1 and 3.3.2.2.1, applicable for train and elevation, hunt shall not exceed 2-1/2 minutes. Any position in train and elevation may be selected for verification of requirements.

3.3.8 Accuracy - With the launcher under control from external dummy director and loaded as specified in 3.3.2.1.1 and 3.3.2.2.1, the error between the director and the power drives shall not exceed the following values:

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CONSTANT VELOCITY	MAXIMUM ALLOWABLE ERROR (minutes)
TRAIN	
1 to 2 degrees per second	3
15 to 20 degrees per second	7
30 to 35 degrees per second	10
ELEVATION	
1 to 2 degrees per second	3
10 to 15 degrees per second	7
SIMPLE HARMONIC MOTION	
TRAIN	
5 degree roll per 9 second period	6
30 degree roll per 9 second period	12
17 degree roll per 4.5 second period	25
ELEVATION	
5 degree roll per 9 second period	6
20 degree roll per 9 second period	10
5 degree roll per 4.5 second period	12

3.3.9 Synchronism firing interlock - The synchronism firing interlocks shall prevent firing when the error between the director and the power drive is more than 20 minutes, plus or minus 10 percent, in either train or elevation.

3.3.10 Synchronization - The launcher shall synchronize to any position order in train within 170 degrees of initial position, providing the limit stops do not interfere, and to any position order in elevation regardless of initial position. Synchronizing time shall be measured from the instant the order is applied to the regulator to the time when the system is, and remains, synchronized to the new order. Synchronism shall be completed within the limits specified below. The conditions of loading shall be as specified in 3.3.2.1.1 and 3.3.2.2.1.

STATIONARY SIGNALS	DISPLACEMENT FROM MIDPOINT OF ROLL	TIME (seconds)
TRAIN		
170 degrees displacement	Not applicable	7-1/2
90 degrees displacement	Not applicable	6
10 degrees displacement	Not applicable	3

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STATIONARY SIGNALS	DISPLACEMENT FROM MIDPOINT OF ROLL	TIME (seconds)
ELEVATION		
60 degrees displacement	Not applicable	5
10 degrees displacement	Not applicable	3
SIMPLE HARMONIC MOTION SIGNALS		
TRAIN		
15 degree roll per 9 second period	18 degrees	7
5 degree roll per 4.5 second period	10 degrees	6
ELEVATION		
15 degree roll per 9 second period	18 degrees	6
5 degree roll per 4.5 second period	10 degrees	6

3.3.11 Return to stow position - The elevation power drive shall synchronize from an elevated position to the stow position within 5 seconds with not more than one overshoot of a 30 minute maximum error when the positions are 60 degrees apart. The guide shall be loaded as specified in 3.3.2.2.1.

3.3.12 Main power failure - With the launcher at maximum velocity in either train or elevation, the power drive affected shall decelerate that applicable portion of the launcher to rest at decelerations not in excess of the following:

TRAIN: 150 degrees per second per second
ELEVATION: 250 degrees per second per second

Requirement applies for a launcher loaded as specified in 3.3.3.1.2 and 3.3.3.2.2, train and elevation, respectively.

3.3.13 Synchro power failure - With the launcher at maximum velocity in either train or elevation, the power drive affected shall decelerate that applicable portion of the launcher to rest at decelerations not in excess of the following:

TRAIN: 150 degrees per second per second
ELEVATION: 250 degrees per second per second

Requirement applies for a launcher loaded as specified in 3.3.3.1.2 and 3.3.3.2.2, train and elevation, respectively.

3.3.14 Train centering pin - The train centering pin, when engaged, shall lock the carriage in the stow position. With the train centering pin handle in the SECURE position, an interlock switch shall prevent starting of the train power drive motor.

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3.3.15 Elevation and depression buffers - With a guide loaded as specified in 3.3.2.2.2, the elevation and depression buffers shall decelerate the guide, from maximum velocity to rest, at a deceleration not in excess of 500 degrees per second per second.

3.3.16 Train buffers - With the guides stowed and loaded as specified in 3.3.2.1.2, the train buffers shall decelerate the launcher from maximum velocity to rest, at deceleration not in excess of 500 degrees per second per second.

3.3.17 Power-off brakes - With the guides loaded and positioned as specified in 3.3.2.1.1, the train power-off brake shall secure the launcher at any position where decelerated to rest by the train power drive or so positioned by the air drive motor. With both cells loaded as specified in 3.3.2.2.1, the elevation power-off brake shall secure a guide at any position where decelerated to rest by the elevation power drive or so positioned by the air drive motor.

3.3.18 Firing interrupter - The firing interrupter shall open the firing circuit to any rail when that rail is positioned within a predetermined non-firing zone. The firing interrupter has two cams. Looking forward with respect to the fore-to-aft centerline of the launcher, the left cam shall interrupt firing on rails No. 1 through No. 4. The right cam shall interrupt firing on rails No. 5 through No. 8. The firing cutout and indicating light switches, when actuated by the switch cams, shall interrupt their respective circuits within 2.0 degrees.

3.3.19 Guide drive pins and elevation stow latches - The hydraulically operated guide drive pins and elevation stow latches shall function as follows:

(a) The elevation stow latches shall secure all guides in the stow position.
(b) The elevation stow latch shall release the guide when the actuating piston is within 1 inch, plus or minus 1/8 inch, of the end of upper stroke.
(c) The elevation stow latch shall be relieved from striking the cross pin in the guide by 1/16 inch, plus or minus 1/32 inch, when the actuating piston is at the bottom of the stroke.
(d) The interlocking circuitry shall permit engagement of one guide drive pin to the torque tube only when all other guides are secured in the stow position.
(e) Time of actuation shall be as specified in 3.3.22.

3.3.20 Missile restraining latch -

3.3.20.1 Positive restraint - With the locking pins in the locked position, the restraining latch shall resist forward and aft motion of the restraining lug, Drawing 1806247, when a minimum force of 20,000 pounds is applied at and parallel with the mounting face of the lug.

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3.3.20.2 Firing detent - With the locking pins in the unlocked position, the restraining latch shall release the restraining lug, Drawing 1806247, when a force of 1,600 to 2,400 pounds is applied in the forward direction. The force shall be applied at and parallel with the mounting face of the restraining lug. The nominal release point shall be 2,000 pounds, plus or minus 400 pounds. With the locking pins in the unlocked position, the restraining latch shall resist aft motion of the restraining lug, Drawing 1806247, when a minimum force of 20,000 pounds is applied at and parallel with the mounting face of the lug.

3.3.20.3 Detent retract mechanism - The detent retract mechanism shall depress the cams of the restraining latch 3/8 inch, plus or minus 1/16 inch, when actuated in the installed position, Drawing 1735447.

3.3.20.4 Restraining latch position - When assembled in guide, at the level indicated on Drawing 1735447, the restraining latch shall mate with the restraining lug of a missile when the missile is so positioned that the vertical face of the detent in the lug is 65.250 inches plus or minus 0.327 inch forward of the machined surface of the aft guide casting, and the top surface of the restraining lug is 15.750 inches plus 0.140 inch minus 0.070 inch below the missile bearing surface of the launching rail.

3.3.21 Missile snubbers - The missile snubbers shall be positioned to conform with Drawing 1735447. Time required for extension and retraction shall be as specified in 3.3.22.

3.3.22 Time sequence of operation - The launcher shall complete each of the automatic steps of operation within the maximum values specified. (Requirements apply for cells loaded with test rounds, Drawing 1453351.)

Sequential Step	Time in seconds (Maximum allowable)
1. Time for guide to go from 4,700 minutes to stow, guide drive pins shift, and next guide to synchronize to 4,700 minutes.	10
2. Time allowed for actuating AUXILIARY FIRING (3A2-S9) to STANDBY.	2.0
3. Time measured from RAIL AND DOOR CONTROL (3A2-S3) turned to EXTEND until EXTENDED lamp illuminated.	3.00 Min. 4.00 Max.
4. Time allowed to actuate AUXILIARY FIRING (3A2-S9) from STANDBY to FIRE.	1.0

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Sequential Step	Time in seconds (Maximum allowable)
5. Time measured from LATCH AND SNUBBER CONTROL (3A2-S4) turned to RETRACT until RETRACTED lamp illuminated.	2.0
6. Time allowed for Relay Transmitter to set up missile.	0.50
7. Time allowed for missile to leave rail.	0.375
8. Time allowed to actuate AUXILIARY FIRING (3A2-S9) to OFF.	1.00
9. Time allowed for time delay relay (3A2-K17) to drop out.	1.25 ±0.75
10. Time allowed to depress Rail Selector push button for new guide selection.	1.00

3.3.23 Control panel indications - All light indications shown on the control panels shall correspond to the actual position and conditions of the launching group.

3.3.24 Launcher Captain's Control Panel correspondence -

3.3.24.1 Dial readings - The observed position dial readings at the Launcher Captain's Control Panel Mk 199 Mod 3 shall correspond with the receiver-regulator synchro dial readings within 2.5 degrees, independent of hunt.

3.3.24.2 Local controls - Each local control (train or elevation) on the Launcher Captain's Control Panel, when released, shall de-energize the stroke control solenoid valve and the transfer valve of its respective power drive.

3.3.24.3 Velocity of the launcher - The velocity of the launcher shall be proportional to any given deflection of the local control in both train and elevation. With the handle deflected fully in either direction, the local control gain shall be capable of adjustment for launcher velocity of 10 degrees per second.

3.3.25 Rail alignment - Each launching rail, when extended, shall meet the following requirements:

(a) With guide engaged to torque tube at 2,000 minutes in elevation, each launching rail shall be parallel to the plane of the train roller path within ±0.007 in 12 inches for any angle of launcher rotation within range specified in 3.3.1.

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(b) With guide engaged to torque tube, each launching rail shall be parallel to a plane which is perpendicular to the plane of the train roller path and the axis of rotation in elevation within 0.010 in 12 inches for any angle of guide elevation within range specified in 3.3.1.

3.3.26 Air drive motor - The air drive motors, train and elevation, shall move the applicable components to any position within the limits imposed by the positive stops when all cells are loaded with test rounds, Drawing 1453351. Velocity in train mode shall be 90 degrees per minute (minimum) and the velocity in elevation mode shall be 10 degrees per minute (minimum) when operating on air pressure at 100 pounds per square inch (psi), plus or minus 10 psi, (measured at the inlet of the launcher hose loop), and control valve handle at maximum displacement. The requirement for velocity in elevation mode applies for a guide being moved between 6,980 minutes and 4,700 minutes.

3.3.27 Dielectric strength and insulation resistance -

3.3.27.1 Dielectric strength - Circuits in Launching Group Mk 16 Mod 4, except those of the Power Drive Amplifiers, Launcher and Missile Simulator Mk 6 Mod 2, loss of pressure switches located in Launcher Mk 112 Mod 2 train and elevation power drives, and direct current (d.c.) tach generators located in train and elevation receiver regulators, shall withstand 720 +0 volts -20 volts alternating current (a.c.) root mean square (rms), 60-cycle high potential, applied between the circuits and structure of each component for one (1) minute without breakdown.

3.3.27.2 Insulation resistance - Circuits in Launching Group Mk 16 Mod 4 less the power drive amplifiers and Missile Simulator Mk 6 Mod 2, shall show an insulation resistance as listed below, when a potential of 500 volts d.c. is applied between the circuits and ground.

Launcher Mk 112 Mod 2	
Stand Mk 107 Mod 2	1 megohm
Carriage Mk 7 Mod 2	1 megohm
Guide Mk 7 Mod 1	3 megohms
Launcher Captain's Control Panel Mk 199 Mod 3	3 megohms

3.3.28 Weather seal requirements -

3.3.28.1 Guide Mark 7 Mod 1 - The guide, when fully assembled at the level indicated on Drawing 1877495, and in a manner capable of stowing and protecting two missiles, shall not permit leakage of water into either cell, which

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either (1) makes contact with enclosed devices in such a manner as to result in failure of operation, or (2) is accumulated in pockets in the bottom or in any part of the guide in such a manner as to result in failure of operation.

3.3.28.2 Carriage Mark 7 Mod 2 - The carriage, when fully assembled at the level indicated by Drawing 1877495, shall not permit leakage of water which either (1) makes contact with enclosed devices in such a manner as to result in failure of operation, or (2) is accumulated in pockets in the bottom or in any other part of the carriage in such a manner as to result in failure of operation.

3.4 Clean-up - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.5 Workmanship - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection -

4.2.1 Individual - Unless otherwise specified, each launcher produced shall be presented for performance testing and inspection for verification of the requirements specified in 3.3.

4.3 Inspection of preparation for delivery - Preparation for delivery of the Launching Group shall be inspected to determine conformance to Section 5 of this specification.

4.4 Performance tests - These tests shall be accomplished on each Launching Group Mk 16 Mod 4 submitted for acceptance under contract. Test methods are detailed in 4.6.

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4.4.1 Test equipment - The following items of test equipment are required to perform the performance tests set forth in this specification:

4.4.1.1 Standard measuring instruments -

(a) One 15-inch precision level, graduated in 0.0005 inch.
(b) Four dial gages, graduated in 0.0001 inch, range to 0.08 inch accurate to one scale division.
(c) One spring scale, graduated in pounds, range one to 100 pounds, accurate to within five percent of scale reading.
(d) One pneumatic pressure gage, graduated in 0.5 pounds, range zero to 50 pounds, accurate to within 1/2 of one percent at any point beyond one pound.
(e) One pneumatic pressure gage, graduated in ten pound intervals, range zero to 800 pounds, accurate within one percent at any point beyond 10 pounds.
(f) One hydraulic pressure gage, graduated in pounds, range zero to 50 pounds, accurate to within 1/2 of one percent at any point beyond one pound.
(g) One hydraulic pressure gage, graduated in 50 pound intervals, range zero to 10,000 pounds, accurate to within one percent at any point beyond one pound.
(h) One pneumatic pressure-relief valve, adjusted to open at 650 pounds per square inch gage (psig).
(i) One pneumatic pressure-relief valve, adjusted to open at 250 psig.
(j) One pneumatic pressure-relief valve, adjusted to open at 150 psig.
(k) One Parker Charging Hose Assembly, Model 1539-602576 (Modified), Drawing 1656700, or equivalent.
(l) One Air Reduction Nitrogen Regulator, Model 11360P, or equivalent.
(m) One torque wrench, cam type, range zero to 50 inch pounds, accurate within 10 percent of scale reading.
(n) One balance scale, flat bed, graduated in pounds, range zero to 1,000 pounds, accurate within 0.5 pound.
(o) One steel-encased thermometer.
(p) One jack, hydraulic or screw, capacity one ton.
(q) One flask, laboratory type, graduated in cubic centimeters (c.c.) range zero to 100 c.c.
(r) One voltmeter, scale range zero to 50 volt, accurate within 5 percent of full scale reading, Simpson Model 260, or equivalent.

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(s) One externally-triggered oscillograph, with at least 25 millivolt per inch vertical sensitivity, Dumont Type 304-A, or equivalent.
(t) Ten rolls oscillograph chart paper, reproducible, longitudinal markings spaced one millimeter, transverse markings spaced five millimeters.
(u) One transformer, step down, primary 110 volts, secondary 6 volts.
(v) Two potentiometers, 2 watts, 10,000 ohms.
(w) Four isolation transformers, high impedance input, 40,000 to 10,000 ohms impedance, Thoardarson Model 20A16, or equivalent.
(x) One circuit analyzer (Dit-MCO, Model 80, or equivalent.

4.4.1.2 Special test equipment (shop accessory) -

(a) One pneumatic shutoff valve, 600 psi.
(b) One hydraulic shutoff valve, 10,000 psi.
(c) Two targets Drawing SA 2372270.
(d) One Boresight Equipment, LD 490381.
(e) One Boresight Telescope Mk 75 Mod 0.
(f) One Rail Gage, LD 497791.
(g) Eight Test Rounds, Drawing 1453351.
(h) One Test Panel, Drawing SA 2373587.
(i) One Dummy Director Mk 3 Mod 8 and Error-Recorder Mk 12 (LD 272569).
(j) One Power Distribution Panel, Drawing SA 2373409.
(k) One Test Set, Drawing SA 2373588.
(l) One Test Fixture, Drawing SA 2369575.
(m) One Tool, Drawing 298871-1.
(n) One Test Fixture, Drawing SA 2357954.
(o) One Hydro Test Fixture, Drawing SA 2357955.
(p) One Test Fixture, Drawing SA 2372138.
(q) One lever, steel, one inch by two inches by 36 inches, with attachment to fit brake drive shaft spline.
(r) One lever, steel, 1/2 inch by one inch by 24 inches, with attachment to fit brake drive shaft spline.
(s) Two Test Cams, Drawing 1656467.
(t) Two Test Cams, Drawing SA 2372685.
(u) One High Potential Test Set, Drawing SA 2490553.
(v) One High Potential Test Kit, Drawing SA 2371509.
(w) One Test Synchro, Drawing SA 2373586.

4.4.2 Acceptance procedures -

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4.4.2.1 Performance tests - The supplier shall conduct the performance tests using the test methods specified in 4.6 to assure that the Launching Group Mk 16 Mod 4 is in compliance with the requirements of 3.3.

4.4.2.2 Test and inspection equipment and facilities - The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel for performing all performance tests. The test equipment shall be of sufficient accuracy and quality to permit performance of the required performance tests.

4.4.2.3 Tests and recording of data - The tests specified herein shall be performed, using the test procedures in Factory Acceptance Test, NAVORD OD 13040. Data observed shall be recorded in the applicable tables and charts provided therein.

4.4.2.4 Miscellaneous inspection - The supplier shall perform or conduct such visual inspection, dimensional checks or other necessary checks to assure that the Launching Group Mk 16 Mod 4 is in compliance with 3.2, 3.3, 3.4, and 3.5.

4.4.2.5 Parts and assembly inspection - Inspection of parts, components and assemblies, and assembling, shall be conducted in accordance with the procedures outlined in NAVORD OD 13040, Section 2, Inspection of Parts, Components and Assemblies, and Section 3, Inspection During Assembling Process.

4.4.2.6 Assembled launching group inspection - Inspection of the assembled Launching Group Mk 16 Mod 4 shall be conducted in accordance with the procedures outlined in NAVORD OD 13040, Section 4, Inspection of Assembled Launching Group Mk 16 Mod 4.

4.5 Test conditions - Unless otherwise specified, the Launching Group Mk 16 Mod 4 shall be subjected to the performance tests under the following conditions:

4.5.1 Temperature - Room ambient 55 degrees F to 115 degrees F.

4.5.2 Altitude - Normal ground atmospheric pressure.

4.5.3 Humidity - Room ambient to 95 percent relative maximum.

4.5.4 Mounting -

4.5.4.1 The Launcher Mk 112 Mod 2 shall be mounted on a test stand which simulates, as far as practicable, the actual foundation. The bottom surface of the stationary mounting ring shall be parallel to a true horizontal plane, within 0.0005 inch in 12 inches.

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4.5.4.2 The Launcher Captain's Control Panel Mk 199 Mod 3 shall be mounted on a suitable stand and shall be connected with the launcher in accordance with the connection list given in Appendix A of NAVORD OD 13040.

4.5.5 Test panel and recording equipment - Unless otherwise specified, the test panel and recording equipment illustrated in NAVORD OD 13040 shall be used in verifying and recording test data.

4.6 Test methods -

4.6.1 Limits of operation - When driven by the air drive motor, the launcher shall meet the requirements of 3.3.1.

4.6.2 Velocity - Operate the launcher under control from external dummy director, Mk 3 Mod 8, in both train and elevation. The launcher shall attain the lowest maximum values, and shall not exceed the maximum values specified in 3.3.2. The velocities shall be recorded by means of a direct recording oscillograph.

4.6.3 Acceleration and deceleration - Launcher performance, for this phase of testing, shall be determined by derivation from the velocity traces obtained during the velocity tests indicated in 4.6.2. The values derived shall meet the requirements of 3.3.3. (Procedures for deriving acceleration and deceleration values are contained in NAVORD OD 13040 and shall be used for obtaining data indicated herein.)

4.6.4 Mechanical-hydraulic balance - Voltage across terminals of torque motor, train and elevation, shall be measured with Launcher operating as specified in 3.3.4.

4.6.5 Mechanical limit stops - The launcher shall be driven into the mechanical limit stops, both train and elevation, at maximum velocity. The velocities shall be recorded by recording oscillograph. Deceleration data, derived from the velocity traces, shall conform to the requirements of 3.3.5.

4.6.6 Static accuracy of correspondence - Error traces shall be run for train and elevation and shall be compared with the director synchro dials. Launcher position, as read from the trace, shall correspond with the dial reading as specified in 3.3.6.

4.6.7 Hunt - Requirement specified in 3.3.7 shall be verified by introducing a static signal of any magnitude within limits, train and elevation, and recording error trace. Trace shall indicate that hunt is within limits specified.

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4.6.8 Accuracy - Launcher accuracy shall be determined by introducing the constant velocity and simple harmonic signals listed in 3.3.8, and recording the error between the input signal and launcher position. The error shall not exceed the maximum allowable, as specified in 3.3.8.

4.6.9 Synchronism firing interlock - The synchronism firing interlock relays in the amplifiers, both train and elevation, shall interrupt the firing circuit when the error between the input signal and either the train or elevation power drive is greater than 20 minutes, plus or minus 10 percent. Requirements shall be verified by recording the opening and closing of the interlock relay when the fine speed synchro of the dummy director is rotated to throw the synchro system out of synchronism with the launcher position. The tests shall be accomplished for both increasing and decreasing orders. The data obtained from the traces shall conform with the requirements specified in 3.3.9.

4.6.10 Synchronization - Operate the launcher under control from dummy director, as described in NAVORD OD 13040. The launcher shall synchronize to the orders listed in 3.3.10. An error trace shall be run for each order specified. The time to synchronize shall be in accordance with the maximum allowable.

4.6.11 Return to stow position - Operate the launcher under control from external dummy director and record an error trace. Position a guide at 5,600 minutes and introduce an order signal to return to stow. Time, overshoot, and maximum error shall be as specified in 3.3.11.

4.6.12 Main power failure - Operate the launcher under control from external dummy director and record velocity. Interrupt main power at maximum velocity. Record main power interruption on velocity trace. Decelerations as derived from the velocity trace shall meet the requirements specified in 3.3.12. These tests apply for train and elevation power drives.

4.6.13 Synchro power failure - Operate the launcher under control from external dummy director and record velocity. Interrupt synchro power at maximum velocity. Record synchro power interruption on velocity trace. Decelerations as derived from the velocity traces shall meet the requirements specified in 3.3.13. Test applies for train and elevation power drives.

4.6.14 Train centering pin - With centering pin extended, use air drive motor to apply torque through train drive gearing to training circle. Torque shall be applied in both directions of train. The carriage shall lock in the stow position without visible backlash. The train power drive shall not start when the train centering pin handle is in the SECURE position.

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4.6.15 Elevation and depression buffers - With the mechanical limit stops in the receiver-regulator of the elevation power drive rendered inoperative, and under conditions of loading specified in 3.3.2.2.2, the guide component shall be driven into the buffers at maximum velocity. A velocity trace shall be recorded. The deceleration value derived from the velocity shall meet the requirements of 3.3.15.

4.6.16 Train buffers - With the mechanical limit stops in the receiver-regulator of the train power drive rendered inoperative, and under conditions of loading specified in 3.3.2.1.2, the rotating component shall be driven into the positive stops at maximum velocity. A velocity trace shall be recorded. The deceleration value derived from the velocity trace shall meet the requirements of 3.3.16.

4.6.17 Power-off brakes - Inspection after installation on the launcher is limited to visual observation for launcher movement after the brake sets, leakage, and release of brake after starting power drive. The brakes, both train and elevation, shall secure their respective component at any position to which the component was decelerated to rest by the power drive, or so positioned by air drive motors. There shall be no external signs of leakage from the lubricating oil lamp. There shall be no sign of leakage at the drain hole for the piston rod.

4.6.18 Firing interrupter - The firing interrupter shall be tested for accuracy by using two test cams, Drawing SA 2372685. The cams shall be installed in the firing interrupter and the launcher shall be moved in train and elevation through the reference points established for cam scribing and machining. Actuation of the firing circuit and indicator light switches shall be within the limits specified in 3.3.18. Results of test shall be logged in NAVORD OD 13040.

4.6.19 Guide drive pins and elevation stow latches - Visual inspection and measurement required for verification that drive pin and stow latch linkage is adjusted as specified in 3.3.19 for (a), (b), and (c). Adjusting to the measurements specified prevents release of guide before drive pin engagement and vice versa. Procedures for adjusting shall be in accordance with OP 2385, Volume 1. Interlock circuit requirement shall be verified as follows:

(a) With the guide component under power, select either rail in Guide No. 1 from Launcher Captain's Control Panel.
(b) Elevate guide to any desired position.
(c) Actuate any other rail button to select a rail in any other guide.
(d) Observe that Guide No. 1 must return to stow before engagement of guide selected in Step c, is accomplished.
(e) Repeat the preceding steps for any combination of rail selection in which transfer from one guide to another is involved. (Actuation of the rail selector button for the other rail within guide transfers missile orders, but has no effect on the guide.)

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4.6.20 Missile restraining latch - Each restraining latch shall be subjected to the following tests before installation in the cells.

4.6.20.1 Positive restraint - Latch shall be installed in Assembly and Test Fixture, Drawing SA 2369575. The locking pins shall be in the locked position. Hydraulic pressure, 1,020 psig, plus 50 psig minus 0, shall be applied to the base end of the cylinder. Pressure shall be sustained for a period of one (1) minute. There shall be no yielding or permanent deformation of parts and components. Tests shall be performed a minimum of five times in this direction. Results shall be recorded in NAVORD OD 13040. Hydraulic pressure, 1,150 psig, plus 50 psig minus 0, shall be applied to the rod end of the cylinder. Pressure shall be sustained for a period of one (1) minute. There shall be no yielding or deformation of parts and components. Tests shall be performed a minimum of five times in this direction. Results shall be recorded in NAVORD OD 13040.

4.6.20.2 Firing detent - With latch installed in Assembly and Test Fixture, Drawing SA 2369575, unlock the locking pins and apply hydraulic pressure, 102 psig, maximum to base end of cylinder. Observe gage and note a sudden drop in pressure when reading has reached the range indicated in the foregoing sentence. Test shall be performed a minimum of five times. Results shall be recorded in NAVORD OD 13040. Hydraulic pressure, 1,150 psig, plus 50 psig minus 0, shall be applied to the rod end of the cylinder. Pressure shall be sustained for a period of one (1) minute. There shall be no yielding or deformation of parts and components. Tests shall be performed a minimum of five times in this direction. Results shall be recorded in NAVORD OD 13040.

4.6.20.3 Detent retract mechanism - Actuate retracting levers and measure full travel of cams. Measurement shall be as specified in 3.3.20.3.

4.6.20.4 Restraining latch position - Using loading procedure detailed in NAVORD OD 13040 load all cells with test rounds, Drawing 1453351. Verify that the vertical face of the lower cam of the restraining latch is in contact with the vertical face of the detent in the restraining lug. Measure the distance from the scale-stop on the test round to a straight edge laid across the machined surface of the aft guide casting. The straight edge and the point at which the measurement is taken shall be within 1/4 inch of the same horizontal plane and vertical plane as the scale-stop. Measurement shall be within the tolerance specified in 3.3.20.4. (The scale-stop is positioned in the same vertical plane as the vertical face of the detent in the restraining lug.) The mating characteristic of the restraining latch with respect to the distance from the missile bearing surface of the launching rail is verified by acceptance of the restraining lug on the test round by the restraining latch: the lug on the test round is within the tolerance specified on Drawing 1806247.

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4.6.21 Missile snubbers - See Drawing 1735447. Using test round, Drawing 1453351, extend snubbers and inspect at all points of contact of snubber shoes with missile.

4.6.22 Time sequence of operation - The sequence of operation, as specified in 3.3.22, shall be performed with the launcher under control from an external dummy director. All functions which specify a time requirement shall be suitably connected with a direct recording oscillograph. The time required for each function to be completed shall conform to those specified in 3.3.22: the sum of time periods recorded for individual launcher actuations plus the allowed time for manual actuations shall not exceed 30 seconds.

4.6.23 Control panel indications - During the examinations and tests indicated herein, the control panels shall be observed and correlated with the known input signals and the readings of the synchro dials in the receiver-regulator of the power drives. The indications observed at the control panel shall correspond to the actual position and condition of the launcher within the limits specified.

4.6.24 Launcher Captain's Control Panel correspondence -

4.6.24.1 Dial readings - Visual observation is required for determining accuracy of correspondence between receiver-regulator synchro dials and the Launcher Captain's Control Panel Mk 199 Mod 3.

4.6.24.2 Local controls - Operate the local controls and observe that launcher, both the train and elevation, comes to rest when the local control is released and returns to STOP position.

4.6.24.3 Velocity of the launcher - Operate local control switch and record velocity trace. Values read from velocity trace shall conform to requirements indicated in 3.3.24.3.

4.6.25 Rail alignment - Each launching rail shall be aligned in accordance with the procedure detailed in NAVORD OD 13040 and shall meet the requirements of 3.3.25.

4.6.26 Air drive motor - With all cells loaded with test rounds, Drawing 1453351, operate the Launcher in train and elevation by means of the air drive motors. Each air drive motor shall carry its applicable assembly to any and all positions within the limits imposed by the positive stops as specified in 3.3.1. Movement may be observed at receiver-regulator synchro dials and time recorded by stop watch. Results shall be logged in NAVORD OD 13040.

4.6.27 Dielectric strength and insulation resistance - All circuits specified in 3.3.27 shall withstand the voltages indicated. Procedures for accomplishing test shall be in accordance with NAVORD OD 13040 and the results shall be logged therein.

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4.6.28 Weather seal requirements -

4.6.28.1 Guide Mk 7 Mod 1 - The guide shall be tested at the level of assembly and installation as shown on Drawing 1877495. A stream of water shall be directed successively over all surfaces. The stream of water shall have a 35 foot head at a nozzle of one inch diameter, and the distance from the nozzle to the unit being tested shall be 10 feet. The duration of the test shall be five minutes. The requirements of 3.3.28.1 shall be met.

4.6.28.2 Carriage Mk 7 Mod 2 - The carriage shall be tested at the level of assembly specified in 3.3.28.2. The stream of water shall be directed successively over all surfaces. The stream of water shall have a 35 foot head at a nozzle of one inch diameter, and the distance from the nozzle to the unit being tested shall be 10 feet. The duration of the test shall be five minutes. The requirements of 3.3.28.2 shall be met.

4.7 Tests after delivery - The launching group, installed on a ship in accordance with the installation instructions of NAVORD OD 10729 will be subjected to the shipboard checkout and performance tests prescribed therein. Any defects discovered during or after these tests, due to supplier's failure to comply with the drawings or specifications for this equipment, shall be corrected by the supplier at no expense to the Government, provided the supplier is notified of such defects within the guarantee period of the contract.

4.8 Rejection - Failure of the Launching Group Mk 16 Mod 4 to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 The Launching Group Mk 16 Mod 4 shall be preserved, packaged, packed, and marked in accordance with levels A or C of NAVORD OD 13041. (See 6.2.)

6. NOTES

6.1 Intended use - The Launching Group Mk 16 Mod 4, covered by this specification, is intended for use on board Naval vessels to launch the Rocket Thrown Depth Charge Mk 2 and the Rocket Thrown Torpedo Mk 3.

6.2 Ordering data - Procurement documents should specify the following information, together with any options desired by the procuring activity.

(a) Title, number, and date of this specification.

(b) Level of preservation required.

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(c) Tools, repair parts and accessories to be furnished.

(d) Special marking required.

(e) Shipping information.

6.3 Storage surveillance - If the equipment specified herein is not installed within the period indicated for the prescribed level of preservation, the equipment shall be inspected for condition, and necessary re-preservation shall be accomplished in accordance with the procedures referenced in 5.1 of this document.

6.4 Description, operation and maintenance - For description, operation, and maintenance of the Launching Group Mk 16 Mod 4, refer to OP 2385, Volumes 1, 2 and 3.

6.5 Definitions -

6.5.1 Supplier - Any organization under contract, capable of furnishing, or causing to be furnished, the product which meets the requirements of this specification.

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Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-L-23956A(OS) LAUNCHING GROUP MARK 16 MOD 4

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

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